

Question 1: For FastEthernet0, verify the IPv4 address, subnet mask and default gateway using ipconfig. (10 pts)

Question 2: For FastEthernet0, verify the address of the IPv4 DNS server using ipconfig /all (10 pts)

```
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 000B.BED5.5026
    Link-local IPv6 Address.....: FE80::20B:BEFF:FED5:5026
    IPv6 Address.....: 2001:DB8:ACAD:1::50
    IPv4 Address.....: 192.168.1.50
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: FE80::1
                           192.168.1.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-D7-8D-61-EC-00-0B-BE-D5-50-26
    DNS Servers.....: ::
                           192.168.1.50
```

Question 3: Verify connectivity to the DNS server by pinging the IPv4 address of the DNS server. (10 pts)

```
C:\>ping 192.168.1.50

Pinging 192.168.1.50 with 32 bytes of data:

Reply from 192.168.1.50: bytes=32 time=2ms TTL=128
Reply from 192.168.1.50: bytes=32 time<1ms TTL=128
Reply from 192.168.1.50: bytes=32 time=12ms TTL=128
Reply from 192.168.1.50: bytes=32 time=11ms TTL=128

Ping statistics for 192.168.1.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 6ms
```

Question 4: Verify connectivity to the default gateway by pinging the IPv4 address of the default gateway (10 pts)

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Question 5: Verify connectivity to the web server by pinging the IPv4 address of the web server (5pts)

```
C:\>ping 10.0.0.99

Pinging 10.0.0.99 with 32 bytes of data:

Reply from 10.0.0.99: bytes=32 time=3ms TTL=128
Reply from 10.0.0.99: bytes=32 time=8ms TTL=128
Reply from 10.0.0.99: bytes=32 time=6ms TTL=128
Reply from 10.0.0.99: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 4ms
```

Question 6: Verify that the DNS server resolves the domain name (URL) that you configured on the DNS server by pinging the URL. (5pts)

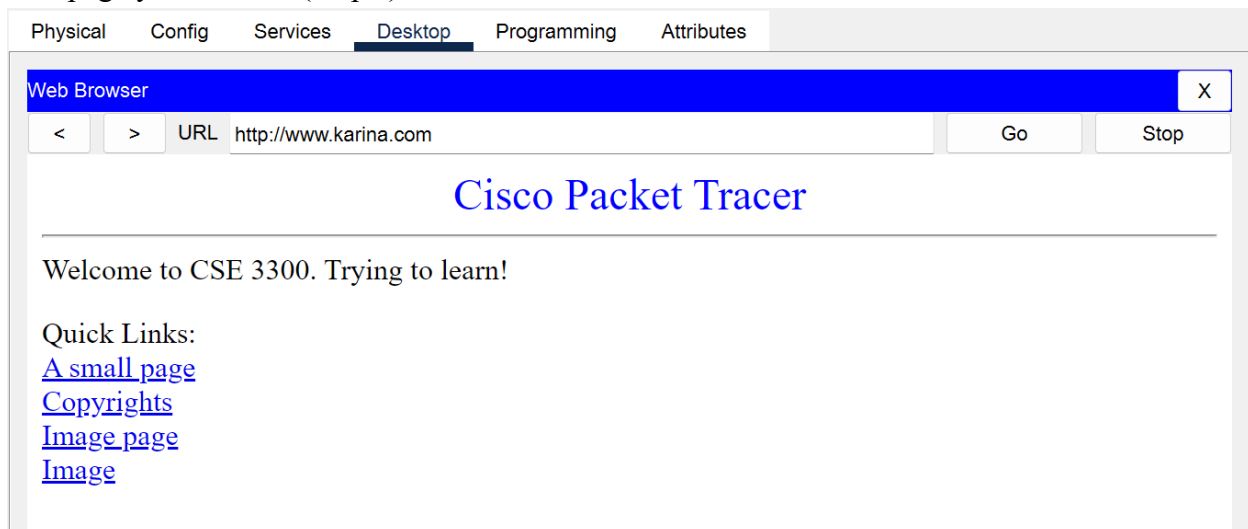
```
C:\>ping www.karina.com

Pinging 10.0.0.99 with 32 bytes of data:

Reply from 10.0.0.99: bytes=32 time=1ms TTL=127
Reply from 10.0.0.99: bytes=32 time<1ms TTL=127
Reply from 10.0.0.99: bytes=32 time<1ms TTL=127
Reply from 10.0.0.99: bytes=32 time=1ms TTL=127

Ping statistics for 10.0.0.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Question 7: Verify that you can now access the HTTP service on the web server and the name is resolved by the DNS server. Upload an image of using the Desktop > Web Browser to access the web page you created. (10 pts)



Question 8: What is the IPv4 network address for PC0 through PC3? Include the subnet mask or prefix length. (10 pts)

IPv4 Addresses: 192.168.1.100/24

Question 9: What is the IPv4 address of the default gateway used by devices PC0 through PC3? (10 pts)

192.168.1.1

Question 10: When the user on PC1 types in www.YourURL.com, which device does PC1 ask for the IPv4 address associated with that URL? (10 pts)

it asks Server1 (192.168.1.50) for the address since Server1 is acting as the DNS server

Question 11: How did PC1 learn the IPv4 address of the DNS server? (10 pts)

it received the information from Server1 through DHCP